

Both transmission channels modulated by $(1 - \varepsilon \bar{P})$

SD layer (information, behaviour)

saliency: $S(t) = \min(1, (I/N) \alpha e^{-\lambda t/100})$

risk: $R_i(t) = (1 - \eta) R_{\text{target}} + \eta R_i(t - 1)$

protection: $P_i = \sigma(\kappa(R_i - \mu))$, $\bar{P} = \text{mean}_i P_i$

modulation

human: $\beta(1 - \varepsilon \bar{P}) I/N$

mosquito: $\text{risk}_m(1 - \varepsilon \bar{P})$

saliency $S(t)$
risk targets ↓

aggregate (S, E, I, R)
mean protection $\bar{P}$ ↑

ABM layer (individuals)

agents with states $S/E/I/R$

stochastic transitions; $\bar{P}$ enters both channels

infected $I(t)$ ↓

risk_m ↑ admissions ↓

beds / ICU ↑

Mosquito layer

humans $\rightarrow E_m \rightarrow I_m \rightarrow$ humans

incubation 10 d, lifespan 14 d

risk_m = $b p_{mh} (I_m/M)$

DES layer (hospital)

beds / ICU / testing queues

deferred admissions (counted once)

capacity constraints